RUT Geometry–Light–Time Coupling Attempt v3

Eikonal Light-Flow Derivation and Late-Time Bridge Mapping

Status: Technical derivation attempt. This note is not a proof of Reverse Universe Theory. It is a controlled attempt to convert the geometry–light–time coupling idea into a field-theory-style scaffold with an explicit light-flow variable, a recoverable GR/ΛCDM limit, and a direct mapping to the late-time bridge behavior seen in the BAO stress tests.

Executive Summary

The v2 coupling note identified the central unresolved issue: the mixed geometry–light–time term could be organized as an effective field theory target, but the light-flow vector L^μ still needed a real definition and dynamics.

This v3 attempt chooses the most conservative next interpretation:

$$L_\mu = \nabla_\mu \theta$$

where θ is an eikonal phase field describing light-front, radiation-front, or null-propagation structure. The mixed scalar becomes

$$Q = L^\mu \nabla_\mu \omega = \nabla^\mu \theta \nabla_\mu \omega.$$

The bridge coupling is then written as

$$\mathcal{L}_{\omega\theta} = -\frac{1}{2}\xi B(a)\left(\nabla^\mu \theta \nabla_\mu \omega\right)^2.$$

At the homogeneous FRW level, this produces an effective kinetic shift in the ω sector:

$$K_{\text{eff}}(a,\omega) = K(\omega) + \alpha B(a),$$

with α defined according to the sign convention of the action. The resulting background equation gives a mathematically clear route from a mixed geometry–light–time term to a late-time $H(z)$ bridge:

$$H^2_{\text{RUT}}(a) = H^2_{\Lambda\text{CDM}}(a) + \frac{1}{3M^2_{\text{Pl}}}\left[\frac{1}{2}K_{\text{eff}}(a,\omega)\dot{\omega}^2 + V(\omega) + \rho_\theta\right].$$

This is a partial derivation win: the late-time bridge can be represented as an effective kinetic/phase-order correction. The remaining hard problem is the full perturbative and causal analysis of the eikonal/light-flow sector.

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1. Motivation from the Bridge-Rescue Simulations

The latest bridge-rescue tests show a consistent pattern:

1. Enlarging r_d alone creates BAO/Hubble tension.
2. Adding a late-time bridge can preserve the triangle:

$$H_0 \approx 73.5, \qquad r_d \approx 181.7\,\mathrm{Mpc}, \qquad \Omega_m \approx 0.29\text{–}0.31.$$

1. Sharp bridge versions can reduce the worst BAO residual below 2σ, but they risk overfitting.
2. Smoother bridge versions survive but reintroduce a mid-redshift residual near $z \sim 0.8$.

Therefore, the theory-side question is:

$$\textbf{Can a physically motivated field coupling generate a late-time/mid-redshift bridge naturally?}$$

This v3 note attempts the first derivation path.

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2. Why the Free Null-Vector Version Was Not Enough

The earlier version used a light-flow vector L^μ with

$$Q = L^\mu \nabla_\mu \omega$$

and a null condition

$$g_{\mu\nu} L^\mu L^\nu = 0.$$

The mixed term was

$$\mathcal{L}_{\omega L} = -\frac{1}{2} Q^2.$$

However, if L^μ is varied as a free vector with only a null constraint, the field equation schematically gives

$$-\xi B Q \nabla_\mu \omega + 2\lambda_L L_\mu = 0.$$

Contracting with L^μ gives

$$-\xi B Q^2 = 0,$$

because $L^\mu L_\mu = 0$. Unless $\xi B = 0$, this forces

$$Q = 0.$$

That would kill the intended bridge.

Therefore, the free-vector version is too under-defined. The light-flow object cannot be only an arbitrary null vector. It must either be derived from a wavefront/eikonal field, treated as an averaged radiation current with its own matter sector, or promoted to a full vector-tensor model with nontrivial dynamics.

This v3 chooses the eikonal route first because it is the least speculative and closest to wave propagation.

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3. Eikonal Definition of Light Flow

Introduce a scalar eikonal phase field θ, whose level surfaces represent wavefronts. Define

$$L_\mu \equiv \nabla_\mu \theta.$$

Then

$$L^\mu = g^{\mu\nu} \nabla_\nu \theta.$$

The null/eikonal condition is

$$g^{\mu\nu} \nabla_\mu \theta \nabla_\nu \theta = 0.$$

The RUT mixed scalar becomes

$$Q = \nabla^\mu \theta \nabla_\mu \omega.$$

Interpretation:

$$Q = \text{light-flow sampling of the phase-order gradient.}$$

When $Q > 0$, light-flow samples increasing ω. When $Q < 0$, light-flow samples decreasing ω. When $Q = 0$, light-flow is tangent to a constant-ω surface.

This gives a clean mathematical marker for phase boundaries:

$$Q = 0.$$

Candidate RUT meanings include horizon-like transition layers, bounce interfaces, phase-state boundaries, or redshift-localized bridge surfaces.

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4. Proposed v3 Action

Use metric signature

$$(-, +, +, +).$$

The v3 action is

$$S=\int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R+\mathcal{L}_m-\frac{1}{2}K(\omega)\nabla_\mu\omega\nabla^\mu\omega-V(\omega)+\mathcal{L}_\theta-\frac{1}{2}\xi B(\chi)\left(\nabla^\mu\theta\nabla_\mu\omega\right)^2+\lambda_\theta\nabla_\mu\theta\nabla^\mu\theta\right].$$

Here:

$$\omega=\text{RUT phase-order / cosmic-breath field},$$

$$\theta=\text{eikonal light-flow phase},$$

$$Q=\nabla^\mu\theta\nabla_\mu\omega,$$

$$B(\chi)=\text{bridge activation function},$$

and χ is a clock variable. In pure FRW work, χ can be taken as a, z, or an effective background scalar. For a fully covariant action, B should eventually depend on a scalar clock field rather than directly on a, because a is gauge/background dependent.

A reviewer-safe covariant replacement is:

$$B(a)\quad\longrightarrow\quad B(\chi),$$

where χ is a monotonic cosmological clock field whose background value tracks scale factor or redshift.

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5. Sign Convention and the Effective Kinetic Shift

The mixed term is

$$\mathcal{L}_{\omega\theta}=-\frac{1}{2}\xi BQ^2.$$

In a homogeneous FRW reduction with $\omega=\omega(t)$, the sign of the resulting kinetic shift depends on how the eikonal field is normalized. Therefore, define the phenomenological bridge coefficient α by

$$K_{\rm eff}(a,\omega)=K(\omega)+\alpha B(a).$$

This absorbs sign convention into α. The no-ghost condition becomes

$$K_{\rm eff}(a,\omega)>0.$$

A safe first modeling prior is

$$K(\omega)>0,$$

$$|\alpha B(a)|<K(\omega),$$

or, for a constant baseline K_0,

$$K_0>0,\qquad -0.5K_0\le\alpha\le1.0K_0.$$

This keeps the coupling perturbative and prevents an immediate ghost instability.

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6. Variation with Respect to ω

Define

$$Q = \nabla^\mu \theta \nabla_\mu \omega.$$

The relevant ω-dependent Lagrangian terms are

$$\mathcal{L}_{\omega,\text{total}} = -\frac{1}{2}K(\omega)(\nabla\omega)^2 - V(\omega) - \frac{1}{2}\xi B(\chi)Q^2.$$

The variation gives

$$\nabla_\mu \left[K(\omega)\nabla^\mu\omega + \xi B(\chi)Q\nabla^\mu\theta\right] - \frac{1}{2}K_{,\omega}(\nabla\omega)^2 - V_{,\omega} = 0.$$

This is the first core v3 field equation:

$$\boxed{\nabla_\mu \left[K\nabla^\mu\omega + \xi BQ\nabla^\mu\theta\right] - \frac{1}{2}K_{,\omega}(\nabla\omega)^2 - V_{,\omega} = 0}$$

Interpretation: the ω field is sourced not only by its own potential and kinetic geometry, but also by how light-flow/eikonal gradients sample ω through Q.

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7. Variation with Respect to θ

Ignoring, for the moment, additional $\mathcal{L}_\theta$ terms beyond the null constraint, the θ-dependent variation gives

$$\nabla_\mu \left[\xi B(\chi)Q\nabla^\mu\omega - 2\lambda_\theta\nabla^\mu\theta\right] = 0.$$

So the second core field equation is

$$\boxed{\nabla_\mu \left[\xi BQ\nabla^\mu\omega - 2\lambda_\theta\nabla^\mu\theta\right] = 0}$$

with the null/eikonal constraint

$$\boxed{\nabla_\mu\theta\nabla^\mu\theta = 0}$$

This is better than the free-vector version because L^μ is no longer varied arbitrarily. It is derived from θ, and the resulting equation is a conservation-like eikonal transport equation rather than an algebraic condition forcing $Q = 0$.

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8. Stress-Energy Tensor of the Mixed Term

The mixed Lagrangian is

$$\mathcal{L}_{\omega\theta} = -\frac{1}{2}\xi B Q^2.$$

where

$$Q = g^{\mu\nu}\nabla_\mu\theta\nabla_\nu\omega.$$

The mixed stress-energy tensor is formally

$$T^{(\omega\theta)}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta\left(\sqrt{-g}\mathcal{L}_{\omega\theta}\right)}{\delta g^{\mu\nu}}.$$

Ignoring the metric dependence of $B(\chi)$ for the first background-level pass, the direct contribution is

$$T^{(\omega\theta)}_{\mu\nu} = \xi B Q\left(\nabla_\mu\theta\nabla_\nu\omega + \nabla_\nu\theta\nabla_\mu\omega\right) - \frac{1}{2}g_{\mu\nu}\xi B Q^2.$$

Therefore, the full Einstein equation becomes

$$M_{\rm Pl}^2 G_{\mu\nu} = T^m_{\mu\nu} + T^\omega_{\mu\nu} + T^\theta_{\mu\nu} + T^{(\omega\theta)}_{\mu\nu} + T^{\chi/B}_{\mu\nu}.$$

The additional term $T^{\chi/B}_{\mu\nu}$ appears if B depends on a dynamical clock field χ whose stress-energy is varied consistently. In a first FRW phenomenological reduction, this can be treated as part of the effective background bridge. In a full paper, it must be included explicitly.

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9. Conservation Bookkeeping

Because the full action is diffeomorphism invariant when all fields are varied consistently,

$$\nabla^\mu T^{\rm total}_{\mu\nu} = 0.$$

Individual sectors do not need to be separately conserved. Instead, the mixed term behaves as an exchange channel:

$$\nabla^\mu T^\omega_{\mu\nu} = +J_\nu,$$

$$\nabla^\mu\left(T^\theta_{\mu\nu} + T^{(\omega\theta)}_{\mu\nu} + T^{\chi/B}_{\mu\nu}\right) = -J_\nu.$$

So total conservation is preserved:

$$\nabla^\mu\left[T^\omega_{\mu\nu} + T^\theta_{\mu\nu} + T^{(\omega\theta)}_{\mu\nu} + T^{\chi/B}_{\mu\nu}\right] = 0.$$

Reviewer-safe meaning: RUT is not claiming energy-momentum appears from nowhere. The bridge term transfers stress-energy between phase-order and light-flow/eikonal sectors while preserving the total conservation law.

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10. FRW Reduction

Use a spatially flat FRW metric for the first pass:

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^{\,2}.$$

Let

$$\omega = \omega(t).$$

For a radial eikonal mode, write locally

$$\theta = \theta(t, r),$$

with the null condition

$$-(\partial_t\theta)^2 + a^{-2}(\partial_r\theta)^2 = 0.$$

This implies

$$\partial_t\theta = \pm a^{-1}\partial_r\theta.$$

For a normalized background light-flow sampling, define

$$\mathcal{N}_\theta \equiv \partial_t\theta.$$

Then

$$Q = \nabla^\mu\theta\nabla_\mu\omega = -\mathcal{N}_\theta\dot{\omega}.$$

If $\mathcal{N}_\theta$ is normalized to unity in the background sampling frame,

$$Q^2 \to \dot{\omega}^2.$$

The effective scalar-sector Lagrangian becomes

$$\mathcal{L}_{\text{eff}}^{\text{FRW}} = \frac{1}{2}K_{\text{eff}}(a, \omega)\dot{\omega}^2 - V(\omega),$$

where

$$K_{\text{eff}}(a, \omega) = K(\omega) + \alpha B(a).$$

This is the core bridge derivation.

11. Effective Density and Pressure

The effective phase-sector density and pressure are

$$\rho_\omega^{\text{eff}} = \frac{1}{2}K_{\text{eff}}\dot{\omega}^2 + V(\omega),$$

$$p_\omega^{\text{eff}} = \frac{1}{2}K_{\text{eff}}\dot{\omega}^2 - V(\omega).$$

The equation-of-state contribution is

$$w_\omega^{\text{eff}} = \frac{\frac{1}{2}K_{\text{eff}}\dot{\omega}^2 - V}{\frac{1}{2}K_{\text{eff}}\dot{\omega}^2 + V}.$$

If $B(a)$ changes over a narrow redshift interval, then K_{eff} changes over that interval, producing a localized modification to ρ_ω^{eff}, p_ω^{eff}, w_ω^{eff}, and therefore $H(z)$.

This maps directly onto the late-time bridge idea.

12. Friedmann Equation and Late-Time Bridge

The RUT-modified Friedmann equation becomes

$$H_{\text{RUT}}^2(a) = \frac{1}{3M_{\text{Pl}}^2}\left[\rho_m + \rho_r + \rho_\Lambda + \rho_\theta + \rho_\omega^{\text{eff}}\right] - \frac{k}{a^2}.$$

Substituting ρ_ω^{eff}:

$$H_{\text{RUT}}^2(a) = \frac{1}{3M_{\text{Pl}}^2}\left[\rho_m + \rho_r + \rho_\Lambda + \rho_\theta + \frac{1}{2}\left(K + \alpha B(a)\right)\dot{\omega}^2 + V(\omega)\right] - \frac{k}{a^2}.$$

Relative to ΛCDM, the bridge correction is

$$\Delta H_{\text{bridge}}^2(a) = \frac{1}{3M_{\text{Pl}}^2}\left[\frac{1}{2}\alpha B(a)\dot{\omega}^2 + \Delta V + \rho_\theta\right].$$

If ΔV and ρ_θ are negligible in the first pass, then

$$\Delta H_{\text{bridge}}^2(a) \approx \frac{\alpha B(a)\dot{\omega}^2}{6M_{\text{Pl}}^2}.$$

This gives the simulation bridge a candidate physical meaning:

$$\boxed{\text{The late-time bridge is an effective kinetic/phase-order shift induced by light-flow sampling of } \omega.}$$

13. Scalar Field Equation in FRW

The reduced scalar equation is

$$K_{\text{eff}}\ddot{\omega} + 3HK_{\text{eff}}\dot{\omega} + \frac{1}{2}K_{\text{eff},\omega}\dot{\omega}^2 + K_{\text{eff},a}aH\dot{\omega} + V_{,\omega} = 0.$$

Since

$$K_{\text{eff}} = K(\omega) + \alpha B(a),$$

we have

$$K_{\text{eff},a} = \alpha B_{,a}.$$

Therefore the bridge contributes the additional driving/friction term

$$\alpha B_{,a}aH\dot{\omega}.$$

This is important: the bridge does not merely add a static energy term. It can actively shape the evolution of ω over the redshift interval where $B(a)$ changes.

A localized bridge around $z \sim 0.8$ can therefore arise from a localized derivative

$$B_{,a} \;/\; 0 \quad =$$

near that epoch.

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14. BAO and Ripple-Horizon Mapping

The sound/ripple horizon can be written schematically as

$$r_d^{\text{RUT}} = \int_{z_d}^{\infty} \frac{c_{\text{eff}}(z)}{H_{\text{RUT}}(z)}\,dz.$$

The v3 bridge affects this through two possible routes:

Route A: Background expansion route

$$K_{\text{eff}} \to H_{\text{RUT}}(z) \to r_d^{\text{RUT}} \to D_M/r_d,\ D_H/r_d.$$

Route B: Effective transport route

$$Q = L^{\mu}\nabla_{\mu}\omega \to c_{\text{eff}}(z) \to r_d^{\text{RUT}}.$$

The conservative first test should use Route A only. Route B should be treated as a later, more speculative extension unless a clean propagation-speed model is derived.

15. Direct Mapping to the Simulation Bridge Function

If the fitted simulation uses a bridge form such as

$$B(a) = \frac{1}{2}\left[1 + \tanh\left(\frac{a - a_*}{\Delta a}\right)\right],$$

then the derivative is

$$B_{,a} = \frac{1}{2\Delta a}\,\mathrm{sech}^2\left(\frac{a - a_*}{\Delta a}\right).$$

This peaks at

$$a = a_*.$$

That gives a mathematical interpretation of the bridge center:

$$\boxed{a_* = \text{epoch where phase-order/light-flow exchange is strongest.}}$$

The width Δa becomes

$$\boxed{\Delta a = \text{duration of the phase-order transition.}}$$

If the best fit demands extremely small Δa, reviewers will suspect overfitting or an unphysical sharp transition. If the bridge survives larger Δa, it becomes more physically credible.

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16. Stability Conditions

16.1 No-ghost condition

$$K_{\text{eff}}(a, \omega) > 0.$$

16.2 Perturbative coupling condition

$$\left|\frac{\alpha B(a)}{K(\omega)}\right| < 1.$$

16.3 Conservative smoothness condition

$$\Delta a \not\to 0.$$

or equivalently in redshift language:

$$\Delta z \gtrsim 0.03\text{–}0.05$$

for the first reviewer-safe stress test.

16.4 Propagation caution

Until the characteristic structure is derived, impose

$$0 \le \left| \frac{\alpha B(a)}{K_{\text{eff}}} \right| \ll 1.$$

This does not prove causal safety, but it avoids immediately large propagation distortions.

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17. GR / ΛCDM Correspondence Limit

The bridge must have a kill switch.

If

$$B(a) \to 0,$$

then

$$K_{\text{eff}} \to K(\omega).$$

If, additionally,

$$\dot{\omega} \to 0,$$

$$V(\omega) \to 0,$$

$$\rho_\theta \to 0,$$

then

$$H^2_{\text{RUT}} \to H^2_{\Lambda\text{CDM}}.$$

So the standard limit is

$$\boxed{B \to 0,\ \dot{\omega} \to 0,\ V \to 0,\ \rho_\theta \to 0 \quad \Rightarrow \quad \text{GR}/\Lambda\text{CDM correspondence.}}$$

This is essential. Without this limit, RUT would conflict with regimes where GR and standard cosmology already work well.

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18. Observable Consequence Chain

The v3 derivation produces a direct test chain:

$$\theta \to L_\mu = \nabla_\mu \theta$$

$$Q = \nabla^\mu \theta \nabla_\mu \omega$$

$$K_{\text{eff}} = K + \alpha B(a)$$

$$H_{\text{RUT}}(z)$$

$$r_d^{\text{RUT}} = \int \frac{c_{\text{eff}}}{H_{\text{RUT}}} dz$$

$$D_M/r_d, \quad D_H/r_d, \quad H_0, \quad f\sigma_8, \quad D_{\Delta t}, \quad \Delta\phi_{\text{GW}}$$

The first serious test remains BAO + covariance. The next tier is BAO + growth/lensing/chronometers/time-delay distances.

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19. Gravitational-Wave Diagnostic

A cautious wave-propagation diagnostic is

$$\Delta\phi_{\text{GW}} \sim \int \xi B(\chi) Q \, d\lambda.$$

In the eikonal interpretation,

$$Q = \nabla^\mu \theta \nabla_\mu \omega.$$

This should not yet be claimed as a waveform model. It is a diagnostic target:

$$\boxed{\text{If gravitational waves sample the same phase-order structure, accumulated phase or siren distance may constrain}}$$

Null test:

$$\Delta\phi_{\text{GW}} \to 0$$

across relevant baselines would force

$$\xi B Q \approx 0$$

in the wave-propagation regime.

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20. What This v3 Attempt Derives

This attempt successfully derives:

1. A non-arbitrary light-flow definition:

$$L_\mu = \nabla_\mu \theta.$$

1. A mixed scalar:

$$Q = \nabla^\mu \theta \nabla_\mu \omega.$$

1. A field-theory-style mixed term:

$$\mathcal{L}_{\omega\theta} = -\frac{1}{2}\xi B Q^2.$$

1. The ω field equation:

$$\nabla_\mu \left[K\nabla^\mu \omega + \xi B Q \nabla^\mu \theta\right] - \frac{1}{2}K_{,\omega}(\nabla\omega)^2 - V_{,\omega} = 0.$$

1. The θ transport equation:

$$\nabla_\mu \left[\xi B Q \nabla^\mu \omega - 2\lambda_\theta \nabla^\mu \theta\right] = 0.$$

1. The FRW effective kinetic bridge:

$$K_{\text{eff}} = K + \alpha B(a).$$

1. The RUT-modified background expansion:

$$H^2_{\text{RUT}} = H^2_{\Lambda\text{CDM}} + \frac{1}{3M^2_{\text{Pl}}}\left[\frac{1}{2}\alpha B(a)\dot{\omega}^2 + \Delta V + \rho_\theta\right].$$

1. The GR/ΛCDM recovery limit.

2. A direct bridge-to-BAO mapping.

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21. What Is Not Fully Derived Yet

This is not yet a complete theory because the following remain unresolved:

1. The exact physical identity of θ: photon eikonal, gravitational-wave eikonal, averaged radiation phase, or effective light-cone structure.
2. The full characteristic analysis proving causal propagation.
3. The full perturbation theory for scalar, vector, and tensor modes.
4. The complete covariance-aware comparison to BAO, growth, weak lensing, cosmic chronometers, and time-delay distances.
5. The physical derivation of $B(\chi)$ rather than fitting it phenomenologically.
6. The full treatment of the clock field χ, if B is made covariant.

These unresolved items should be treated as the v4/v5 theory-hardening agenda, not hidden.

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22. Reviewer-Safe Claim

A reviewer-safe summary is:

> We show that the RUT late-time bridge can be represented at the homogeneous background level as an effective kinetic shift in a phase-order field ω, induced by a mixed eikonal light-flow coupling $Q = \nabla^{\mu}\theta\nabla_{\mu}\omega$. This produces a mathematically controlled route from geometry–light–time coupling to a localized modification of $H(z)$, while preserving a GR/ΛCDM correspondence limit when the bridge decouples. The result does not prove RUT, but it supplies a concrete field-theory target for testing whether the phenomenological BAO bridge can be derived rather than merely fitted.

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23. Bottom Line

The v3 derivation attempt is a meaningful advance.

It does not fully prove the geometry–light–time coupling, but it converts the idea into a testable scaffold:

$$\boxed{\theta \to L_{\mu} = \nabla_{\mu}\theta \to Q = \nabla^{\mu}\theta\nabla_{\mu}\omega \to K_{\text{eff}} = K + \alpha B(a) \to H_{\text{RUT}}(z) \to r_d/BAO/H_0.}$$

The strongest derived result is:

$$\boxed{\text{The late-time bridge can be interpreted as a temporary effective kinetic shift in the RUT phase-order sector.}}$$

The key next question is:

$$\boxed{\text{Can } B(\chi) \text{ be derived from the dynamics of } \omega, \theta, \text{ and/or a covariant clock field instead of inserted by hand?}}$$

Suggested filename:

RUT_Geometry_Light_Time_Coupling_Attempt_v3_Eikonal_Light_Flow_Derivation.pdf